

V⁺ ionisation limit 54411.67 cm⁻¹ = 6.74619 eV (183.784 nm)

Arrow colour = vacuum wavelength (nm)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e J J $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4H_{132} - z^4F_{15Q}$	445.3255	445.2005	673.1985	7.516	8.100e+07	12.94	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

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^{51}V hyperfine structure ($I = 7/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$y_{F3D_2}^0$	993(0)	F=1: -1112.98 F=2: -915.12 F=3: -618.32 F=4: -222.60 F=5: +272.06 F=6: +865.65	
$y_{F3D_2}^1$	154(0)	F=1: -168.63 F=2: -138.65 F=3: -93.69 F=4: -33.73 F=5: +41.22 F=6: +131.16	
$y_{F3D_2}^2$	540(6)	F=2: -3642.48 F=3: -2023.60 F=4: +134.91 F=5: +2833.04	
$x_{F3D_2}^0$	690(6)	F=2: -4384.48 F=3: -3605.00 F=3: -2435.81 F=4: 876.89 F=5: +1071.76 F=6: +3410.14	
$x_{F3D_2}^1$	303(0)	F=1: -6779.06 F=2: -5573.89 F=3: -3766.14 F=4: -1355.51 F=5: +1657.10 F=6: +5272.60	
$x_{F3D_2}^2$	259(0)	F=1: -4508.08 F=2: -3707.99 F=3: -3071.99 F=4: -2835.28 F=5: -1672.09 F=6: +218.10	
$x_{F3D_2}^3$	181(0)	F=1: -3058.63 F=2: -2740.85 F=3: -2264.18 F=4: -1862.62 F=5: -834.17 F=6: +119.17 F=7: +1231.40 F=8: +2550.52	
$x_{F3D_2}^4$	81(0)	F=2: -1841.48 F=3: -1598.64 F=4: -1274.87 F=5: 876.15 F=6: -384.48 F=7: +182.12 F=8: +2550.52	
$w_{F3D_2}^0$	258(30)	F=1: -2900.50 F=2: -2384.85 F=3: -1611.39 F=4: -580.10 F=5: +709.01 F=6: +2255.94	
$w_{F3D_2}^1$	108(30)	F=1: -169.89 F=2: -159.91 F=2: -1376.04 F=3: -1052.27 F=4: -80.94 F=5: +80.94 F=6: +566.61 F=7: +1322.08	
$z_{F3D_2}^0$	183(30)	F=4: -5440.47 F=5: -4526.11 F=6: -3428.87 F=7: -2148.76 F=8: -685.57 F=9: +960.08 F=10: +2788.81 F=11: +4800.42	
$z_{F3D_2}^1$	213(30)	F=2: -1436.76 F=3: -798.20 F=4: +53.21 F=5: +1117.48	
$z_{F3D_2}^2$	144(30)	F=0: -2266.43 F=1: -2122.53 F=2: -1834.72 F=3: -1403.03 F=4: -827.42 F=5: -107.92 F=6: +755.48 F=7: +1762.78	
$u_{F3D_2}^0$	285(30)	F=1: -5482.46 F=2: -4912.85 F=3: -4058.44 F=4: -2919.23 F=5: -1495.22 F=6: +213.60 F=7: +2027.22 F=8: +4485.65	

Measured A, B constants; sources in the data file